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GRAPH NEURAL NETWORKS FOR AUTONOMOUS THEOREM PROVING

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Abstract

The main goal of this work is to prove the existence of.....

Notation: We will work over an algebraically closed field K of characteristic zero.

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Chapter 1

Generalized Algebraic Theories

The goal of this chapter is give meaning to the concept of term, type and theory, while introducing some tools that we will use later when we present the algorithm.

By theories we mean Generalized Algebraic Theories (or GATs), introduced by John Cartmell in his thesis [1] [2]. GATs give us both a very idiomatic way of presenting theories and powerful interpretation tools like Context Categories, introduced later in this section. The notion of generalized algebraic theory is a generalization of the notion of many-sorted algebraic theory in the following manner: while in Martin Löff's many-sorted algebraic theories sorts are constant types, interpreted as sets, in generalized algebraic theories sorts might be constant or variable, over a specified range of types, and variable types are interpreted as families of sets. So a generalized algebraic theory consists of:

1. a set of sorts, that can be constant or variable;
2. a set of operators, with the sort of the arguments and value specified;
3. a set of axioms, where each axiom is an identity between similar expressions.

Let's present category theory for example: The set of sorts contains two elements Ob and Hom. Ob is a constant type and Hom is a variable type, for a pair of terms of type Ob, name them x, y, we have a type $Hom(x, y)$. Here we see a key point in general algebraic theories, types vary over terms, no over types. The set of operators also has two elements id and $\circ$. The id operator has one argument x of type Ob, and returns a value of type $Hom(x, x)$. This shows how the type of value of an operator might vary depending on the arguments. The $\circ$ operator (or composition) expects 5 arguments, three of type Ob, name them x, y, z, one of type $Hom(x, y)$, say f, and one of type $Hom(y, z)$, named g, and it returns a value of type $Hom(x, z)$. Here we see that the type of an argument might also vary

depending on a previous argument. Informally, instead of writing $\circ(x, y, z, f, g)$ we can write $\circ(f, g)$, as the first three arguments might be inferred from the last two. This notation will be discussed in a further section.

Now we will introduce the notation we will be using when talking about terms and types. This notation will be formally defined in a later section. So consider you have a set of variables V, we will associate terms to variables as required. When asserting that a term t is of type Δ we will write $t : \Delta$, but if t has variables $x_1, \ldots, x_n$ occurring in it this assertion does not make sense unless we assign some types to $x_1, \ldots, x_m$. To do this we will write:

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t : \Delta}$$

that reads as: if x_1 is a variable of type Δ_1, ...and if x_n is a variable of type Δ_n, then t is a term of type Δ. Similarly, if we want to assert that if x_1 is a variable of type Δ_1, ...and if x_n is a variable of type Δ_n, then Δ is a type, we can write:

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t : \Delta}$$

this assertions are called rules. Rules are used to express which expressions are well-formed. We will work with this rules as units instead of the base expressions. This is because terms and types do not make sense by themselves but they need to be thought of in a context.

Axioms are also written as rules. The conclusion of those rules are equalities between types or between terms of the same type. We write those rules as:

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta = \Delta'} \qquad \frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t = t' : \Delta}$$

and read as: if x_1 is a variable of type Δ_1, ...and x_n is a variable of type Δ_n, then $\Delta = \Delta'$, or $t = t'$ as terms of type Δ, respectively.

To continue with the previous example, we can write the axioms of category theory as:

$$\begin{aligned}
&x, y : Ob, f : Hom(x, y), \quad \circ(id(x), f) = f \\
&x, y : Ob, f : Hom(x, y), \quad \circ(f, id(y)) = f \\
&w, x, y, z : Ob, f : Hom(w, x), g : Hom(x, y), h : Hom(y, z), \\
&\qquad \circ(\circ(f, g), h) = \circ(f, \circ(g, h))
\end{aligned}$$

When presenting a theory's language i.e. the sets of sorts and operators, we must present each symbol with a rule specifying the role of this symbol, either as a sort symbol or as specifically typed operator symbol. This rules are called introductory rules. In the case of a sort symbol A the rule should be like

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{A(x_1, \ldots, x_n) \text{ is a type}}$$

clearly specifying the types A is dependent on. In the case of an operator symbol, a rule of the form

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{f(x_1, \ldots, x_n) : \Delta}$$

suffices to specify the types of the arguments and the value of f.

Finally, to present a theory we will give a set os symbols, each associated to its introductory rule, and a set of axioms. And, of course, everything must be well-formed, something we will address in the following section. So now we can give a formal presentation of the theory of categories:

Symbol	*Introductory Rule*
Ob	Ob is a type.
Hom	$x, y \in \mathrm{Ob} : \mathrm{Hom}(x, y)$ is a type.
$\circ$	$x, y, z \in \mathrm{Ob}$, $f \in \mathrm{Hom}(x, y)$, $g \in \mathrm{Hom}(y, z) : \circ(f, g) \in \mathrm{Hom}(x, z)$.
id	$x \in \mathrm{Ob} : \mathrm{id}(x) \in \mathrm{Hom}(x, x)$.

Axioms

$\circ(\mathrm{id}(x), f) = f$,	whenever $x, y \in \mathrm{Ob}$ and $f \in \mathrm{Hom}(x, y)$.
$\circ(f, \mathrm{id}(y)) = f$,	whenever $x, y \in \mathrm{Ob}$ and $f \in \mathrm{Hom}(x, y)$.
$\circ(\circ(f, g), h) = \circ(f, \circ(g, h))$,	whenever $w, x, y, z \in \mathrm{Ob}$, $f \in \mathrm{Hom}(w, x)$, $g \in \mathrm{Hom}(x, y)$ and $h \in \mathrm{Hom}(y, z)$.

1.1 Formal Definition

In this section we will formally define all of the concepts we just introduced. To give a notion of well-formed introductory rule, we need the concept of drivability, which also depends on the introductory rules. To tackle this problem we will leave aside the question of well-formedness until we have the complete set of derived rules of a theory. We will first define a pretheory and then define a theory as a well-formed pretheory.

We will assume to have countable set of variables V. Lets define the notion of rule on an alphabet W, the elements of W are called symbols.

Definition 1.1. *The set of* ***expressions*** *is defined inductively such that expressions are finite sequences of elements of $W \cup V \cup \{(\} \cup \{)\} \cup \{,\}$ defined by the clauses:*

1. *if $x \in V$, then x is an expression;*
2. *if $L \in W$, then L is an expression;*
3. *if $L \in W$ and $e_1, \ldots, e_n$ are expressions, then $L(e_1, \ldots, e_n)$ is an expression.*

Now we can define the building blocks of a rule:

Definition 1.2. *A **premise** is defined to be a sequence of $V\times$ the set of expressions. The empty sequence is allowed and we call it an empty premise. If $((x_1, \Delta_1), \ldots, (x_n, \Delta_n))$ is a premise, we write it as $x_1 : \Delta_1, \ldots, x_n : \Delta_n$.*

Definition 1.3. *We call a **conclusion** to one of the following:*

1. *T-conclusion: defined by a single expression Δ and written as 'Δ is a type'.*
2. *$\in$-conclusion: defined by a pair of expressions (t, Δ) and written as '$t \in \Delta$'.*
3. *T =conclusion: defined by a pair of expressions (Δ_1, Δ_2) and written as '$\Delta_1 = \Delta_2$'.*
4. *$\in$=conclusion: defined by a triple of expressions (t_1, t_2, Δ) and written as '$t_1 = t_2 : \Delta$'*

Finally a **rule** is determined by a premise P and a conclusion C, and is written as $\frac{P}{C}$. We name the rule as T-rule, $\in$-rule, T =rule or $\in$=rule according to the form of the conclusion.

Now we can define pretheory and later theory.

Definition 1.4. *A **pretheory** consists of*

1. *a set of sort symbols S;*
2. *a set of operator symbols Z;*
3. *for each sort symbol A, an associated rule of the alphabet $S \cup Z$:*

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{A(x_1, \ldots, A_n) \text{ is a type}}$$

   for some $n \geq 0$. It is called the introductory rule of A;

4. *for each operator symbol F, an associated rule of the alphabet $S \cup Z$:*

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{F(x_1, \ldots, x_n) : \Delta}$$

   for some $n \geq 0$. It is called the introductory rule of F;

5. *a set of axioms. Each axiom is either a T =rule or a $\in$=rule of the alphabet $S \cup Z$.*

Now we will tackle the problem of well-formedness of rules.

Definition 1.5. *If U is a pretheory, then*

1. *a* ***context*** *is a premise* $x_1 : \Delta_1, \ldots, x_n : \Delta_n$ *such that the rule*

$$\frac{x_1 : \Delta_1, \ldots, x_{n-1} : \Delta_{n-1}}{\Delta_n \text{ is a type}}$$

is a derived rule such that x_n *is distinct from all of* $x_1, \ldots, x_n$.

2. (a) *the rule* $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta \text{ is a type}}$ *is* ***well-formed*** *if the premise* $x_1 : \Delta_1, \ldots, x_n : \Delta_n$ *is a context.*

   (b) *the rule* $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t : \Delta}$ *is* ***well-formed*** *if* $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta \text{ is a type}}$ *is a derived rule of* U.

   (c) *the rule* $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta = \Delta'}$ *is* ***well-formed*** *if* $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta \text{ is a type}}$ *and* $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta' \text{ is a type}}$ *are derived rules of* U.

   (d) *the rule* $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t = t' : \Delta}$ *is* ***well-formed*** *if* $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t : \Delta}$ *and* $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t' : \Delta}$ *are derived rules of* U.

Definition 1.6. *The set of* ***derived rules*** *of a pretheory* U *is the set of rules derivable by the following principles of derivation:*

Definition 1.7. *A pretheory is well-formed if all of its introductory rules are well-formed. A* ***theory*** *is a well-formed pretheory.*

1.2 The Substitution Lemma

Given $\Delta, t_1, \ldots, t_n$ expressions and $x_1, \ldots, x_n$ variables, then we define the expression $\Delta[t_1|x_1, \ldots, t_n|x_n]$ the result of simultaneous replacement of $x_1, \ldots, x_n$ in Δ by $t_1, \ldots, t_n$. We call this the Substitution operation.

The substitution lemma states that the set of derived rules of a theory is closed under the operation of substitution of correctly typed terms for variables. We could add substitution as one of the derivation rules but would make applying induction very messy.

Definition 1.8. *A derived rule of* $\dfrac{y_1|\Omega_1, \ldots, y_n|\Omega_n}{Conclusion}$ *of a theory* U *is said to have the* ***substitution property*** *if for all contexts* Q *of* U*, for all expressions* $s_1, \ldots, s_m$ *of* U *if the rules*

$$\frac{Q}{s_1 : \Omega_1}, \frac{Q}{s_2 : \Omega_2[s_1|y_1]}, \ldots, \frac{Q}{s_m : \Omega_m[s_1|y_1, \ldots, s_{m-1}|y_{m-1}]}$$

are all derived rules of U, then the rule

$$\frac{Q}{Conclusion[s_1|y_1, \ldots, s_m|y_m]}$$

is a derived rule of U.

We will see that all derived rules hold this property.

Lemma 1.9. *If* $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{Conclusion}$ *is a derived rule of U, then any variable appearing in the conclusion occurs among $\{x_1, \ldots, x_n\}$.*

Proof. Easily obtained by applying induction over the derivation rules. □

Lemma 1.10.

1. *the premise of a derived rule is a context,*
2. *if $x_1 : \Delta_1, \ldots, x_n : \Delta_n$ is a context, then for all $1 \leq i \leq n$, then* $\dfrac{x_1 : \Delta_1, \ldots, x_{i-1} : \Delta_{i-1}}{\Delta_i \text{ is a type}}$ *is a derived rule.*

Proof. □

Lemma 1.11 (The Substitution Lemma)**.** *Every derived rule of a theory U has the substitution property.*

Proof. □

Corollary 1.12 (Change of Variables)**.** *If* $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{Conclusion}$ *is a derived rule of U and if $y_1, \ldots, y_n$ is a sequence of distinct variables, then*

$$\frac{y_1 : \Delta_1, y_2 : \Delta_2[y_1|x_1], \ldots, y_n : \Delta_n[y_1|x_1, \ldots, y_{n-1}|x_{n-1}]}{Conclusion[y_1|x_1, \ldots, y_n|x_n]}$$

is a derived rule of U.

1.3 Contexts and Realizations

Now we have all the syntax of our theories, we should introduce it's semantics. As mentioned earlier, one of the main benefits of working with GATs is the construction of Context Category, that we will give in this section.

First we will start by defining some concepts key to understanding the structure of this categories.

Our notion of **tree** consist of trees with a unique least element. Let Θ be a tree. If A is a node of Θ we will write $A \in \Theta$, confusing the tree with its set of nodes. Given two nodes $A, B \in \Theta$ such that B succeeds A, we write $A \lhd B$ in Θ. The least node of a tree will be denoted as 1. If we have a node A of Θ different from 1, then there exists a unique $n \geq 0$ and a unique sequence $A_1, \ldots, A_n$ such that $1 \lhd A_1 \lhd A_2 \lhd \cdots \lhd A_n \lhd A$ in Θ.

We call the **tree of families** to the following tree structure over the collection of sets, families of sets, families of families of sets, ... where a set A precedes all families of sets indexed by A, a family of sets B precedes all families of families of sets indexed by B, ... If A_1 is a set, and A_2 is a A_1-indexed family of sets then for $a_1 \in A_1$ we write $A_2(a_1)$ for the value of the family A_2 at a_1. If we have $1 \lhd A_1 \lhd A_2 \lhd \cdots \lhd A_n \lhd A$ in the tree of families, then for all $a_1 \in A_1$, ... and for all $a_n \in A_n(a_1, \ldots, a_{n-1})$, $A(a_1, \ldots, a_n)$ is a set.

Now we will define the category of contexts and realizations of a theory U.

Definition 1.13. *If $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$ and $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle$ are contexts of U, then a **realization** of $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle$ with regard of $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$ is a m-tuple $\langle t_1, \ldots, t_m \rangle$ such that for each j, $1 \leq j \leq m$, the rule*

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t_j : \Omega_j[t_1|y_1, \ldots, t_{j-1}|y_{j-1}]}$$

is a derived rule of U.

Then, for a theory U, we define the category of contexts and realizations of U, $R(U)$, as the category whose objects are contexts of U, and where we have a morphism

$$\langle t_1, \ldots, t_m \rangle : \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle \longrightarrow \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle$$

for each realization $\langle t_1, \ldots, t_m \rangle$ of $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle$ with regard of $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$. The identity of an object $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$ is the realization $\langle x_1, \ldots, x_n \rangle$. And the composition of two morphisms

$$\langle t_1, \ldots, t_m \rangle : \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle \longrightarrow \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle$$

$$\langle s_1, \ldots, s_w \rangle : \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle \longrightarrow \langle z_1 : \Lambda_1, \ldots, z_w : \lambda_w \rangle$$

is defined as

$$\langle s_1, \ldots, s_w \rangle \circ \langle t_1, \ldots, t_m \rangle = \langle s_1[t_1|y_1, \ldots, t_m|y_m], \ldots, s_w[t_1|y_1, \ldots, t_m|y_m] \rangle$$

The set of objects of $R(U)$ i.e. the set of contexts of U is structured as a tree with a single least element $\langle\rangle$, the empty context, and where every context $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n$ is preceded by $x_1 : \Delta_1, \ldots, x_{n-1} : \Delta_{n-1}$. Moreover, for any context $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n, x_{n+1} : \Delta_{n+1}, \ldots, x_{n+m} : \Delta_{n+m} \rangle$ of U, $\langle x_1, \ldots, x_n \rangle$ is a realization of $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$ with respect to $\langle x_1 : \Delta_1, \ldots, x_{n+m} : \Delta_{n+m} \rangle$. We name that morphism as $p(\langle x_1 : \Delta_1, \ldots, x_{n+m} : \Delta_{n+m} \rangle, \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle)$.

Lemma 1.14. *If $A = \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$, $A' = \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle$ and $B = \langle y_1 : \Omega_1, \ldots, y_{m+w} : \Omega_{m+w}$ are contexts, and $f = \langle t_1, \ldots, t_m \rangle$ is a realization of $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle$ with regard to $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$, then*

$$\begin{tikzcd} f^*B \arrow[r, "q(f,B)"] \arrow[d, "p(f^*B,A)"'] & B \arrow[d, "p(B,A')"] \\ A \arrow[r, "f"'] & A' \end{tikzcd}$$

is a pullback diagram in $R(U)$, where

$$f^*B = \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_{m+1} : \Omega_{m+1}[t_1|y_1, \ldots, t_m|y_m], \\ \ldots, y_{m+w} : \Omega_{m+w}[t_1|y_1, \ldots, t_m|y_m] \rangle$$

$$q(f, B) = \langle t_1, \ldots, t_m, y_{m+1}, \ldots, y_{m+w} \rangle$$

This properties motivate the following definition:

Definition 1.15. *A **contextual category** consists of*

1. *a category C with a terminal object 1;*
2. *a tree structure on the objects of C such that the terminal object 1 is the unique least element of the tree;*
3. *for all $A, B \in ObC$ such that $A \lhd B$ a morphism $p(B) : B \to A$ in C and that will be written as $B \twoheadrightarrow A$;*

4. *for all $A, A' \in ObC$ and for all $f : A \to A'$ in C, for all $B \in ObC$ such that $A' \lhd B$, an object f^*B in C and a morphism $q(f, B) : f^*B \to B$ such that the diagram*

$$\begin{tikzcd} f^*B \arrow[r, "q(f,B)"] \arrow[d] & B \arrow[d] \\ A \arrow[r, "f"'] & A' \end{tikzcd}$$

is a pullback in C such that

   (a) *for all $A, B \in ObC$ such that $A \lhd B$, $id_A^*B = B$ and $q(id_A, B) = id_B$;*

   (b) *when*

$$\begin{tikzcd} & & B \arrow[d] \\ A \arrow[r, "f"] & A' \arrow[r, "f'"] & A'' \end{tikzcd}$$

   in C, then $(f' \circ f)^*B = f^*(f'^*B)$ and $q(f' \circ f, B) = q(f', B) \circ q(f, f'^*B)$.

The category $R(U)$ is structured as a contextual category by the tree structure and pullbacks defined earlier.

Chapter 2

The Proposed Model

Consider the presentation of a theory U. The algorithm has one goal, given a derived T-rule $\dfrac{Q}{\Delta \text{ is a type}}$ of the theory U it will try to create a term of type Δ under the context Q, i.e. finding a realization from the context Q to the context (Q, Δ). To do this we will consider the context category $C(U)_Q$, to limit ourselves to reasoning under the context Q.

Let G be a finite subcategory of $C(U)_Q$ containing both Q and (Q, T). We apply iteratively the following steps

1. assign a label to all the objects of G in 3 categories: 'keep', 'expand' and 'delete'. (this is where the AI model comes in)

2. Let G_1 be the category consisting of all objects of G not labeled as 'delete'.

3. For each object X of G labeled as 'expand' apply the following expansion rules:

   (a)

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